



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VI Semester
End Semester Examination - April / May -2025
MACHINE LEARNING [ECE-4409]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Distinguish between univariate and multivariate regression? (4)
 - A)
 - B) Discuss two example applications of machine learning? (3)
 - C) Discuss the concept of hypothesis? How many hypotheses are possible in a two-class classification problem? (3)
- 2) Compute the covariance matrix for the following data (4)

$$X = \begin{bmatrix} 4 & 2 & 2 & 3 & 4 \\ 1 & 4 & 3 & 6 & 4 \end{bmatrix}$$

 - A)
 - B) Explain the computational steps used in LDA with appropriate equations (4)
 - C) Explain SVM for linearly non-separable data (2)
- 3) With a neat schematic, explain the architecture of multilayer perceptron (4)
 - A)
 - B) Explain the properties of impurity measure used in learning decision tree? (4)
 - C) Discuss on posterior probability? (2)
- 4) With a neat schematic explain bootstrapping technique used for evaluating the performance of a classifier? (4)
 - A)
 - B) Discuss the concept of Receiver Operating Characteristics (ROC) for two class classification problem? How area under the curve (AUC) is computed? (4)
 - C) Explain confusion matrix used in evaluation of a classifier? (2)
- 5) Discuss the concept of k-nearest neighbor method used for non-parametric density estimation? (4)
 - A)

- B) Discuss the differences between partitional and hierarchical clustering methods? (3)
- C) Discuss Gaussian mixture model used in clustering? (3)

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